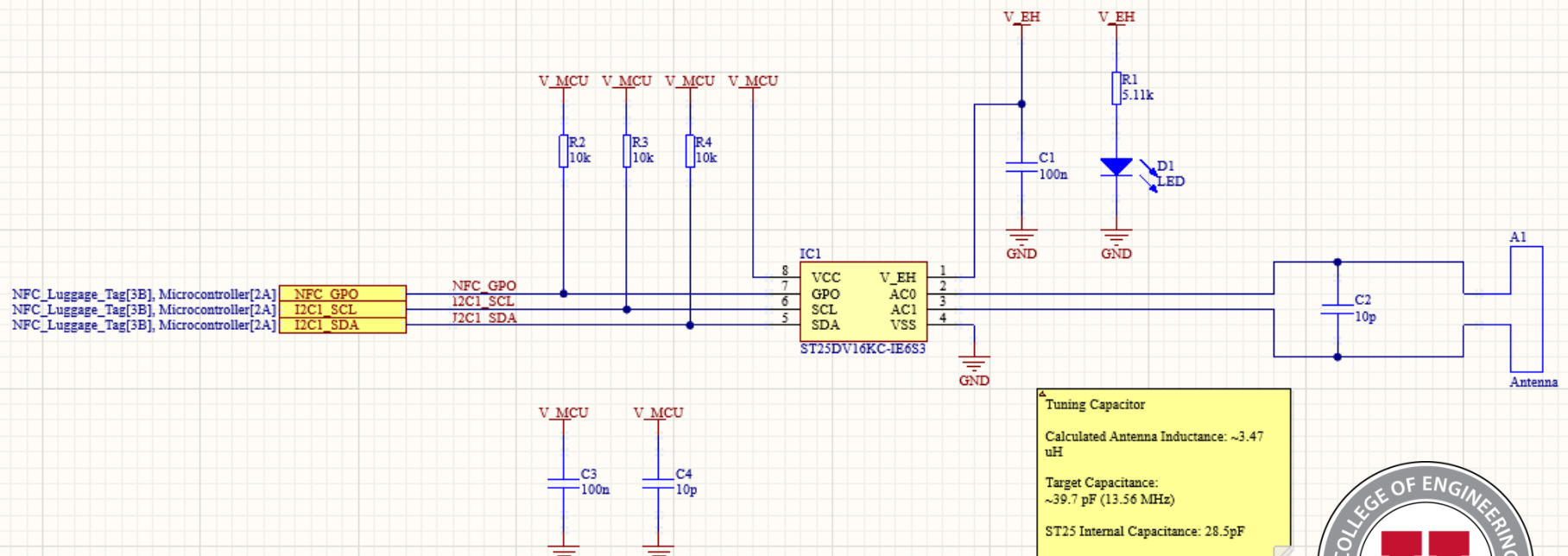


IEEE PCB Design Workshop: (Week 02) Datasheets and Components



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University of Utah, Salt Lake City, UT



Workshop Outline

Tentatively scheduled for 6-7 weeks

- Sept. 15 – Introduction to Schematics (WEB L102)
- **Sept. 22 – Schematics Continued (WEB 112)**
- Sept. 29 – Introduction to PCB Layout (WEB L102)
- Oct. 6 – Layout Continued (WEB L102)
- Oct. 8 – PCBs DUE!
- **** FALL BREAK ****
- TBD – Soldering Week 1
- TBD – Soldering Week 2
- TBD – Soldering Week 3 (Optional)

Workshop Logistics

- We **highly** recommend attending all workshop sessions
 - Each session is designed to build off previously discussed materials
- By the end of the workshop, you will be able to:
 - Build an NFC Luggage Tag
 - Understand the basics of schematic layout, PCB layout, and board manufacturing
 - Gain experience with through-hole and surface mount PCB assembly
 - Have a basic understanding of manufacturing techniques and equipment used in the industry

Announcements

- Alternative Projects
 - We will be able to get alternative project boards manufactured if submitted by the deadline.
 - Limited to the 10 cm x 10 cm dimensions outlined by JLCPCB.
 - Talk or email me if you have any questions!

Want more experience?

- Consider joining the FSAE tractive team!
 - The Tractive Team is currently looking for students to assist with designing and assembling the electrical system for an electric formula-style race car!
 - No experience required!



U of U FSAE Discord Link



Component Selection

- Component selection poses a challenge during the design stage.
- Selection should be driven by design constraints.
 - Cost
 - Size
 - Performance

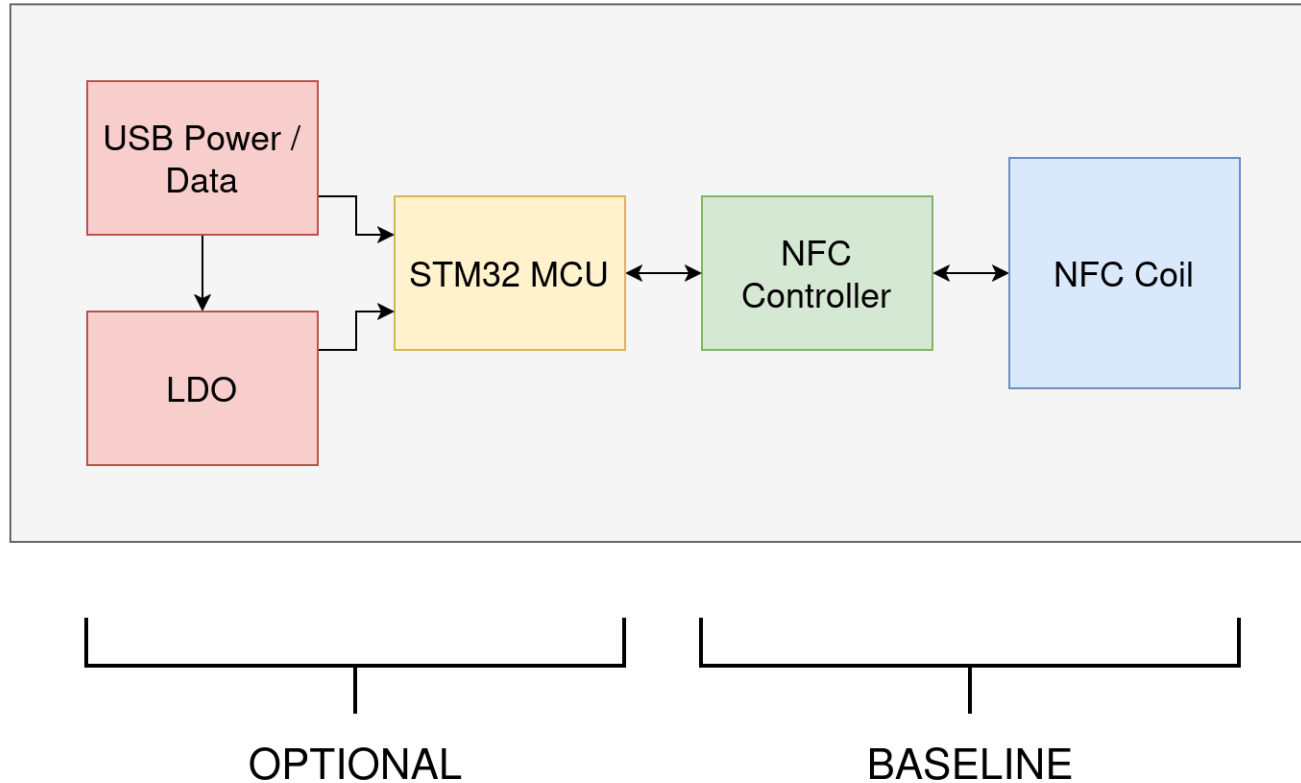
Component Search

- Manufacturers and Distributors may provide component search functions
- Manufacture Search:
 - Specific Functionality, Performance
 - More details on specifics
 - Narrower Scope
- Distributor Search:
 - Wider Component Reach
 - Multiple Manufacturers
 - Broader Search Parameter



DigiKey

Block Diagram



Coil Calculations

- STMicroelectronics has a coil calculation
 - Use the tool!
- Application Notes for calculations as well!
 - ST
 - NXP



Geometry

Turns

Antenna Length mm

Antenna Width mm

Conductor

Width mm

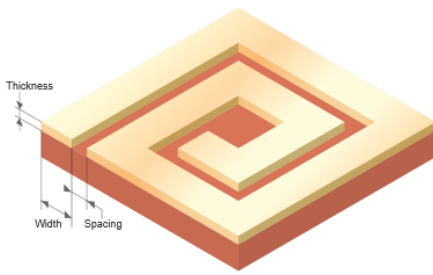
Spacing mm

Thickness μm

Substrate

Thickness mm

Relative Electrical Permittivity (ϵ_r)



Antenna Results

Segmentation Mode: Vertical

Segments: 24

Equivalent Inductance: **3.47 μH**
@13.56 Mhz



Resonance Calculations

- Inductance is estimated
- Use value to calculate the values for the capacitance for resonance at 13.56 MHz

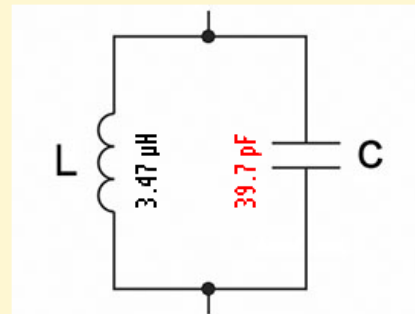
LC Resonant Frequency Calculator

Inductance, L	3.47	μH
Capacitance, C	?	pF
Resonant Frequency, f	13.56	MHz

[Calculate](#) [Reset](#)

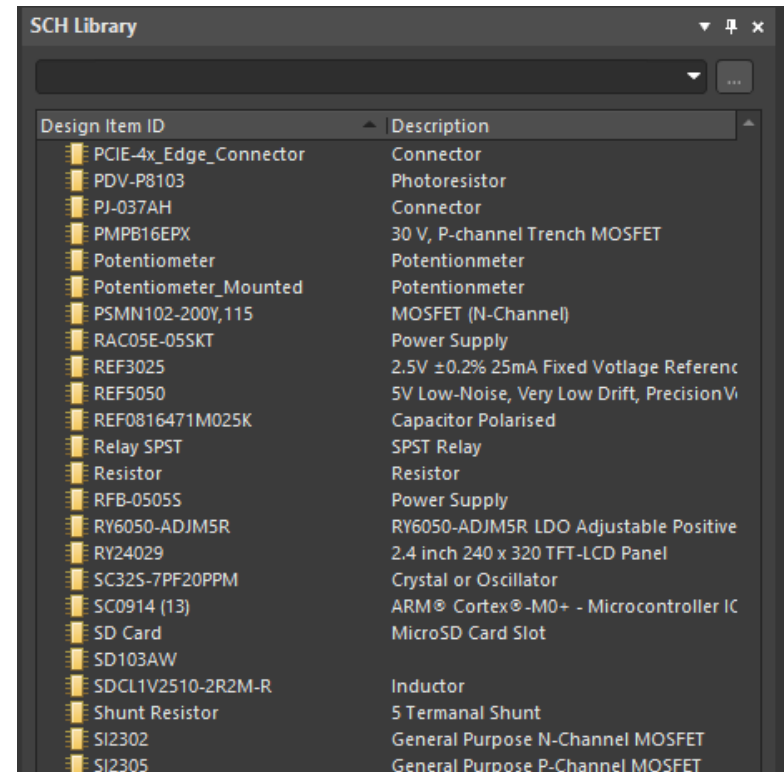
Results

Capacitance (C) = **39.700082515 pF**



Component Libraries

- Libraries are collections of components and footprints
 - All components for a project must be derived from a library
 - Projects may reference multiple different component libraries
 - Relatively consistent across platforms
- Typically, there are symbol and footprint libraries
 - Tightly coupled together

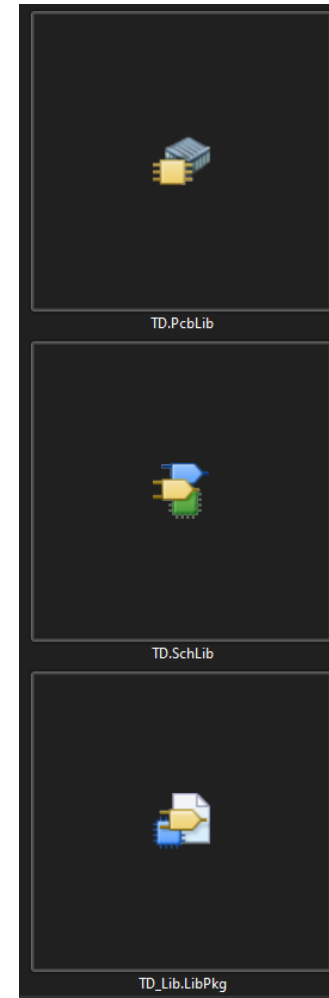


The screenshot shows a software window titled "SCH Library". It contains a table with two columns: "Design Item ID" and "Description". The table lists various electronic components and their specifications.

Design Item ID	Description
PCIE-4x_Edge_Connector	Connector
PDV-P8103	Photoresistor
PJ-037AH	Connector
PMPB16EPX	30 V, P-channel Trench MOSFET
Potentiometer	Potionmeter
Potentiometer_Mounted	Potionmeter
PSMN102-200Y,115	MOSFET (N-Channel)
RAC05E-05SKT	Power Supply
REF3025	2.5V $\pm$ 0.2% 25mA Fixed Voltage Referenc
REF5050	5V Low-Noise, Very Low Drift, Precision V
REF0816471M025K	Capacitor Polarised
Relay SPST	SPST Relay
Resistor	Resistor
RFB-05055	Power Supply
RY6050-ADJM5R	RY6050-ADJM5R LDO Adjustable Positive
RY24029	2.4 inch 240 x 320 TFT-LCD Panel
SC325-7PF20PPM	Crystal or Oscillator
SC0914 (13)	ARM [®] Cortex [®] -M0+ - Microcontroller IC
SD Card	MicroSD Card Slot
SD103AW	
SDCL1V2510-2R2M-R	Inductor
Shunt Resistor	5 Termanal Shunt
SI2302	General Purpose N-Channel MOSFET
SI2305	General Purpose P-Channel MOSFET

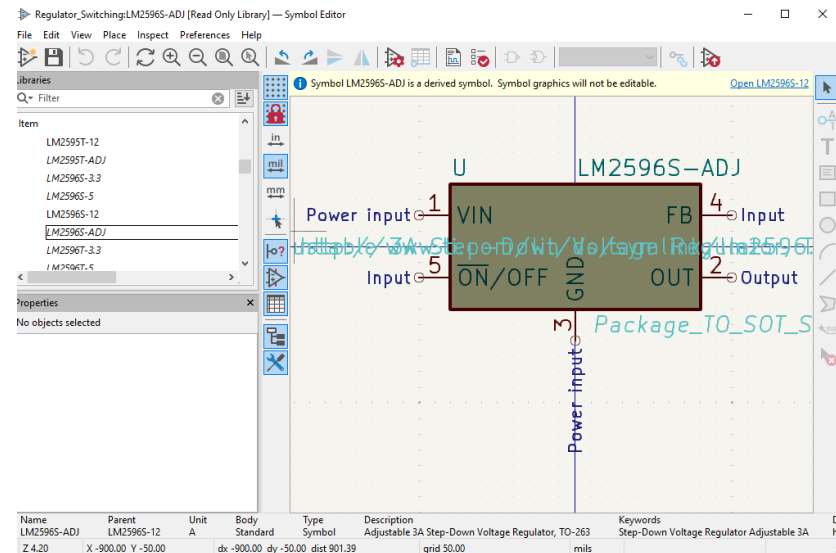
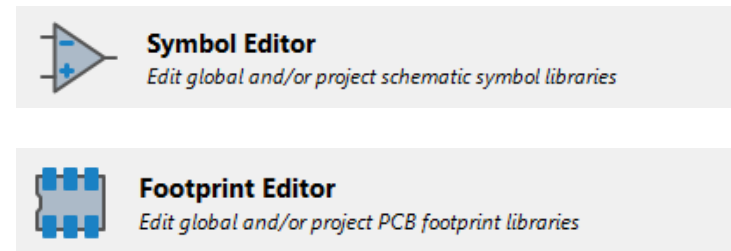
Component Library Types (Altium)

- .SchLib
 - Contains all component schematic symbols
- .PcbLib
 - Contains all footprint information
- .LibPkg
 - Also known as an integrated library
 - Packages multiple .SchLib and .PcbLib libraries together for easier management



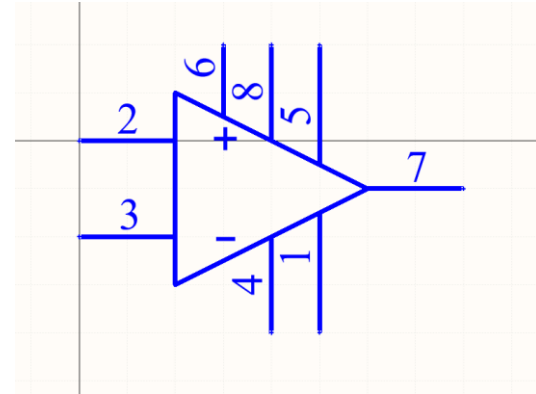
Component Library Types (KiCAD)

- .kicad_sym
 - Contains all the symbol information for components in the library
- .pretty
 - Contains all the footprint information to be used with components



Making Component Symbols

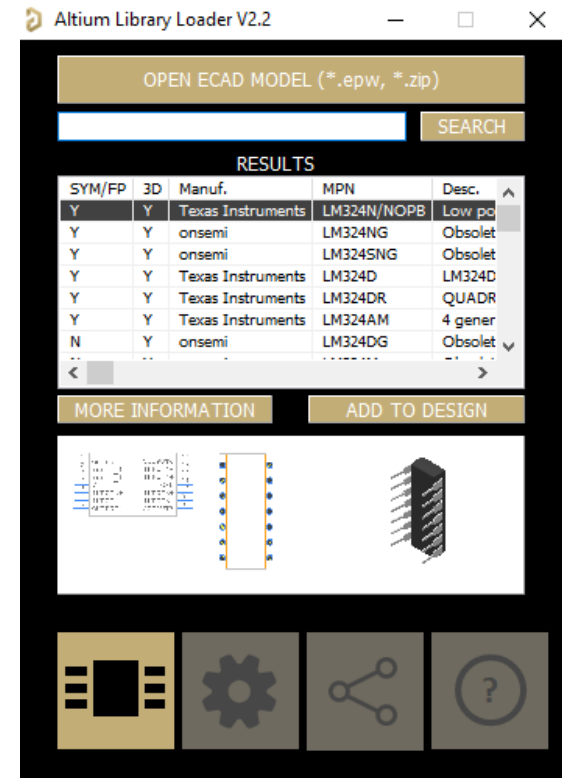
- Making symbols is a time-consuming process
- All pins and designators must be correct, or they will result in errors during layout
- Requires the ability to read a datasheet



5	OUT0	VDD	16
6	OUT1		
7	OUT2	SDI	2
8	OUT3	CLK	3
9	OUT4	LE/DM1	4
10	OUT5	SDO	14
11	OUT6	OE/DM2	13
12	OUT7		
1	GND	R-EXT	15

SamacSys (Component Manager)

- SamacSys is a third-party component manager
 - Integrates with Altium with an extension
 - Creates schematic symbols and footprints for you
 - Can be used with KiCAD as well



Questions?

Questions?

Download Today's Project Files

Navigate to the workshop GitHub and
download today's files

<https://github.com/IEEE-U-of-U/IEEE-PCB-Workshop-Spring-2026>